

NAUTILUS: Navigational Agentic Understanding Through Interpretable Localization Using Streetscapes

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Abstract

Classical image-geolocation systems predict coordinates from visual input but provide little insight into the evidence behind a decision. We study whether state-of-the-art multimodal large language models (MLLMs) can perform geolocation while also explaining their predictions. Across 25 European scenes, we record visual cues, hypotheses, exploration choices, and final pins, adding covered static-image runs to measure the effect of game-rendered road labels. This evaluation reveals failures that coordinate error alone cannot explain, including correct place recognition followed by incorrect map placement and missed evidence caused by limited exploration. For open-weight GLM-5.3-Flash, we assess reasoning through its exposed trace, comparing reported cues with the reasoning and actions behind them. Our results show that MLLMs can combine competitive geolocation with useful, though imperfect, evidence-based explanations, making them a promising direction for more interpretable visual geolocation.

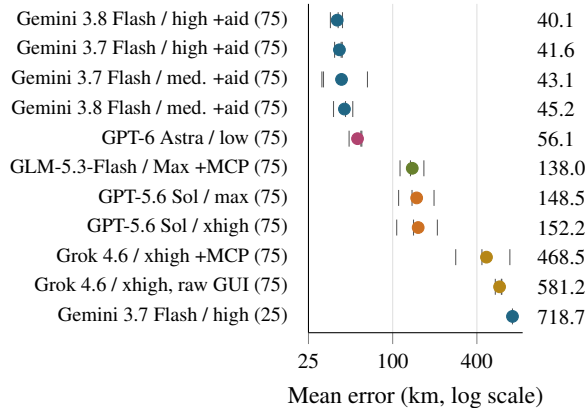


Figure 1. Final-pin distance error. Colored dots show pooled model means; black ticks mark the mean error of each individual run. Parentheses give prediction counts, and “aid” denotes Gemini’s pin helper.

Table 1. Static baselines on 25 images. Errors are in km. A_{200} is the percentage of predictions within 200 km.

Baseline	Mean ↓	Median ↓	A_{200} ↑
PLONK	501.8	167.8	56%
Chipoint v2	536.2	312.1	32%
GeoCLIP	1,183.3	224.2	44%
SALAD + OSV-5M	1,308.5	929.2	20%

1. Introduction

We evaluate state-of-the-art multimodal large language models (MLLMs) available as of September 2026. Prior benchmarks assess geolocation accuracy and reported visual cues [6, 8, 12]. NAUTILUS links each evaluated model’s initial guesses, subsequent observations, and final pins. Repeated runs and covered road-overlay labels reveal how interpretation and control affect predictions. [Our evidence explorer](#) presents the results and saved observations.

2. Method

Scenes and baselines. We curate 25 European OpenGuessr scenes: 8 easy, 9 medium, and 8 hard, according to the specificity of their visual cues. Images and Street View links fix starting views and targets. We evaluate four existing geolocation baselines under the static-image condition (Table 1). Pretrained weights and implementation settings are documented in the source project.

Interactive evaluation. We retain 28 full 25-location evaluations and nine additional Grok MCP games, three per difficulty, totaling 75 rounds. Sol and Astra operate the map visually; Gemini Flash uses pin-placement aid, with an unaided 3.7 control; Grok uses GUI or MCP; and GLM uses MCP in ZCode. Prompts request five initial cues and a geographic belief before exploration and pin refinement. Runs allow about five minutes per round and exclude external lookup, hidden metadata, and page/network inspection.

Covered-image comparison. Fourteen of the 25 scenes display a street name or road number as text rendered by the game. To measure its effect while accounting for variation in

exploration and tool use, we evaluate each scene with three interactive runs and one covered static-image run. Each of the nine MLLM settings therefore contributes 42 interactive predictions and 14 covered-image predictions.

Evidence and metrics. We take distance error from OpenGuessr’s result screen when available and otherwise compute it from WGS84 coordinates; all errors are in km. Cue reports and screenshots let us check the evidence and camera actions reported by every MLLM. GLM’s exposed reasoning trace also lets us compare its report with its reasoning and actions.

3. Results

Localization accuracy. Gemini 3.8 Flash variants with pin-placement aid achieve the lowest mean final-pin errors, at 40.1–45.2 km (Figure 1), with Astra next at 56.1 km. GLM and Sol max/xhigh follow at 138.0, 148.5, and 152.2 km; GLM’s 22.5 km median shows that many predictions are much closer than its mean suggests. Grok with MCP averages 468.5 km, slightly better than PLONK, the strongest static baseline, at 501.8 km (Table 1), while raw-GUI Grok and unaided Gemini trail both. Static baselines generally lag the interactive MLLMs. Part of the gap is practical: MLLMs can read street text and other cues, move to reveal new views, and refine their pins.

Interface and belief errors. Grok shows why map control matters: it recognizes Place de la Bastille, yet its raw-GUI marker lands about 980 km away. Across all 75 MCP-assisted rounds, pins match Grok’s chosen coordinates; MCP corrects placement, not geography. By contrast, the GPT models’ strong computer use keeps pins consistent with their beliefs, even when those beliefs are wrong. Sol (max) mistakes Pula Arena for Rome’s Colosseum and pins Rome accordingly. Astra (low) reads the Lithuanian road number (3426), finds it near Radviliškis on the in-game map, refines along the route, and finishes 85 m away.

Covered versus uncovered inputs. Models usually mention visible road text in their cue reports, although occasional omissions can make its influence easy to underestimate. Covering it makes six of nine settings less accurate (Figure 2); Gemini 3.7 Flash high rises from 21.8 to 185.5 km mean error. Written cues can therefore affect predictions even in the few reports that omit them. The effect varies: Sol xhigh is nearly unchanged (149.8 versus 155.8 km), while GLM improves from 131.3 to 93.9 km and Grok from 594.3 to 281.8 km. Road labels can help or mislead.

Which cues remain useful? The Place de la Bastille scene provides clear evidence that MLLMs can extract useful visual cues: all nine covered-image predictions locate it within 110 m by identifying the July Column and a transit sign. The Greek-road image exposes the limit of less distinctive evidence. All nine covered-image guesses choose another country, with errors of 454–2,331 km, after empha-

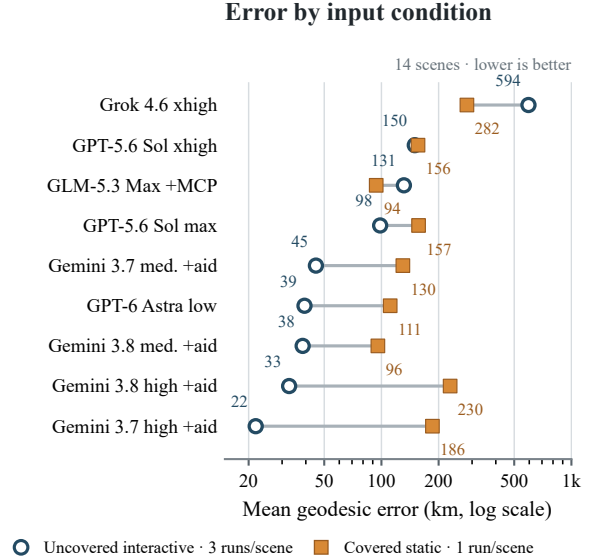


Figure 2. Mean geodesic error for the 14 scenes containing road-overlay text. Uncovered values pool three interactive runs per scene ($n = 3 \times 14$); covered values use one static prediction per scene ($n = 14$).

sizing Mediterranean vegetation and limestone shared across several countries.

Visible text can also trigger wrong associations. GLM interprets the game-rendered “N444” road-overlay label as evidence for the Netherlands in all three runs, whereas its covered-image guess favors Belgium from housing and overhead wiring. Brick architecture likewise shifts its Bologna guess away from two initial Milan hypotheses.

Why repeated runs diverge. On the “N444” scene, moving forward is the decisive action. GLM makes this choice only in Run 2, uncovers additional evidence, revises the Netherlands to Belgium, and finishes 18.2 km away. Runs 1 and 3 remain at the starting view and exceed 136 km. The result therefore depends not only on interpreting visible cues but also on how proactively the model uses MCP navigation to seek new evidence. On the Greek road, an ineffective drag in Run 3 reveals nothing new despite a report of improved legibility, and the run finishes 660.2 km away.

Conclusion. State-of-the-art MLLMs already perform end-to-end image geolocation: they inspect scenes, form hypotheses, and sometimes operate the browser and map without MCP tools. Their explanations add a capability missing from coordinate-only baselines: cited landmarks, road signs, architecture, and vegetation make both successes and failures inspectable. These cues are imperfect, and navigation or pin-placement errors can still separate a correct belief from its submitted coordinate. Evidence and action traces improve interpretability.

A. Related Work

MLLM geolocation benchmarks. LLMGeo shows that fine-tuning improves open models, making them competitive with some closed models at country-level geolocation [12]. IMAGEO-Bench evaluates distance error, geographic bias, and structured reasoning across three image collections [6]. These evaluations motivate our study of contemporary game-playing agents.

Reasoning and cue reliability. EarthWhere scores the use of annotated visual cues alongside country- and street-level answers [8]. GeoRC compares explanations against 800 expert chains for 500 scenes and documents geographic misattribution, hallucinated cues, and claims of unavailable tool use [9]. We connect these claims to recorded camera transitions and submitted pins.

Exploration and tool use. WanderBench supplies navigable panorama graphs, while GeoAoT acquires additional views through planned actions [14]. ERGeoBench compares single-view, panorama, and sequential yaw/pitch/zoom conditions, exposing difficulties in metric localization and cross-view consistency [13]. Meanwhile, LocationAgent combines a Reasoner–Executor–Recorder hierarchy with external tools for evidence verification [7], and Thinking with Map trains agents to query map APIs [5]. NAUTILUS examines camera-control failures and belief-to-pin transfer under restricted external lookup.

Cue masking and faithfulness. IMAGEO-Bench links successful localization to identifiable visual anchors and, in its revised text-masking experiment, measures accuracy losses after removing signage, storefront names, and license-plate text [6, App. J]. Durrieu et al. test attribution-guided object regions through deletion and insertion [3]. Our masks target Google-rendered road labels and provider credits, while traces record remaining cues. Because language-model explanations can omit influences on their answers [10], we interpret those statements together with the observed views, actions, and coordinates.

Specialized reference systems. GeoCLIP aligns images with GPS representations [11], SALAD aggregates features for retrieval [4], and PLONK models geographic uncertainty generatively [2]. Alongside OSV-5M reference imagery [1], these methods provide the static context for our agent-focused analysis.

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